CS 426

Cloud Computing and IoT

Course Details

Cloud Computing and IoT Cloud Computing Fundamentals, Types of cloud, Cloud services: Benefits and challenges of cloud computing, Cluster Computing, Grid Computing, Grid Computing Versus Cloud Computing, Key Characteristics of Cloud Models, Cloud Services and File System, Virtualization: Virtualization of CPU, Memory, I/O Devices, Virtualization for Data-center Automation. IoT and Digitization, IoT Challenges, IoT Network Architecture and Design, Smart Objects: Sensors, Actuators, and Smart Objects, Sensor Networks, Connecting Smart Objects, Communications Criteria, Introduction to Arduino, Fundamentals of Arduino Programming, Introduction to RaspberryPi.

Suggested Readings:

1. Kai Hwang, Geoffrey Fox, Jack Dongarra, Distributed and Cloud Computing, Elsevier, 2012.
2. Rajkumar Buyya, Christian Vecchiola, and Thamarai Selvi, Mastering Cloud Computing, TMH,
3. 2013.
4. Dan C. Marinescu, Cloud Computing: Theory and Practice, Elsevier, 2013.
5. Samuel Greengard, The Internet of Things, MIT Press, 2015
6. Peter Waher, Learning Internet of Things, Packt Publishing, 2015
7. Dirk Slama, Frank Puhlmann, Jim Morrish , Enterprise IOT, O'Reilly Publishers, 2015

Marks Distribution

The marks distribution for the course CS 426 (Cloud Computing) is as follows:

Quiz I-II: 15 Marks
Quiz II/Project: 20 Marks
Assignments/Class Performance: 15 Marks
End Semester: 50 Marks

Self-Reading/Exposure Exercises:

- CloudSim Simulator using Eclipse https://www.youtube.com/watch?v=OZRbkkEuQMI&list=PLGe95EHDPn-X5Opz5xB-3rUnqr_PXOOnv
- Cloud4SciEng.org
- Amazon Web Services (AWS)
- https://www.youtube.com/watch?v=kOjdExBUqAI
- https://opensource.com/article/19/3/resources-raspberry-pi
- Research Papers: Information to be given in the class